

DEVELOPMENT OF AUTONOMOUS WHEEL ROBOT

ANTI SLEEP ALARM



KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

— Deemed to be University U/S 3 of UGC Act, 1956 —

Participants:-

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[Helps drivers to not sleep while driving car preventing accidents and if unconscious then the motor slows down until stopped.]

AIM

To make anti sleep alarm where hobby motor stop running when blinking isn't done for 4 sec and a piezo buzzer is used to create a sound.

COMPONENTS REQUIRED

- ✧ ARDUINO UNO
- ✧ ARDUINO IDE
- ✧ WIRES
- ✧ 9V BATTERY
- ✧ RELAY MODULE
- ✧ HOBBY MOTOR
- ✧ BLINKING SENSOR
- ✧ PEIZO BUZZER

THEORY

-This project is based on the issue of drivers falling asleep while driving for long distances on the road. This project involves a blink sensor consists of an IR transmitter and an IR receiver attached to a transparent spectacles. The transmitter transmits IR rays into the eye. If the eye is shut, the output of the receiver turns HIGH. If the output is high for more than three seconds, a buzzer will go off first to indicate the driver that he had kept his eye shut for three seconds. After five seconds, along with the buzzer, brakes are applied on the wheels of the motor, thereby bringing the vehicle to a stop. The relay module is used in this project to act as a switch to avoid the circuit to burn.

PROGRAM

```
const int blinkPin = 2;
const int motorPin = 13;
const int buzzerPin = 12;

long time;

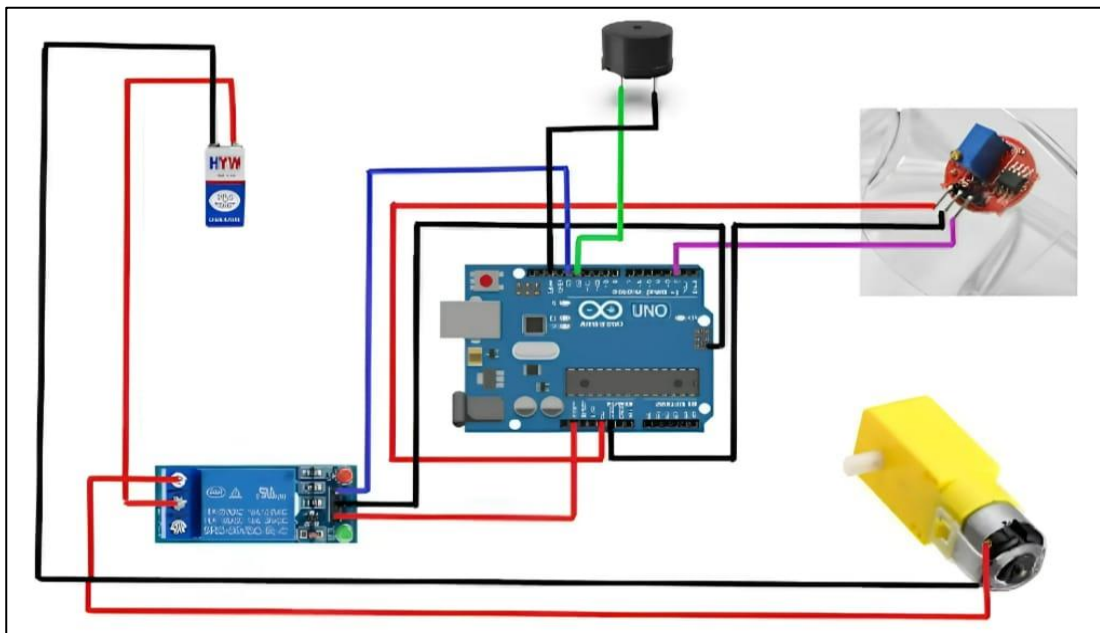
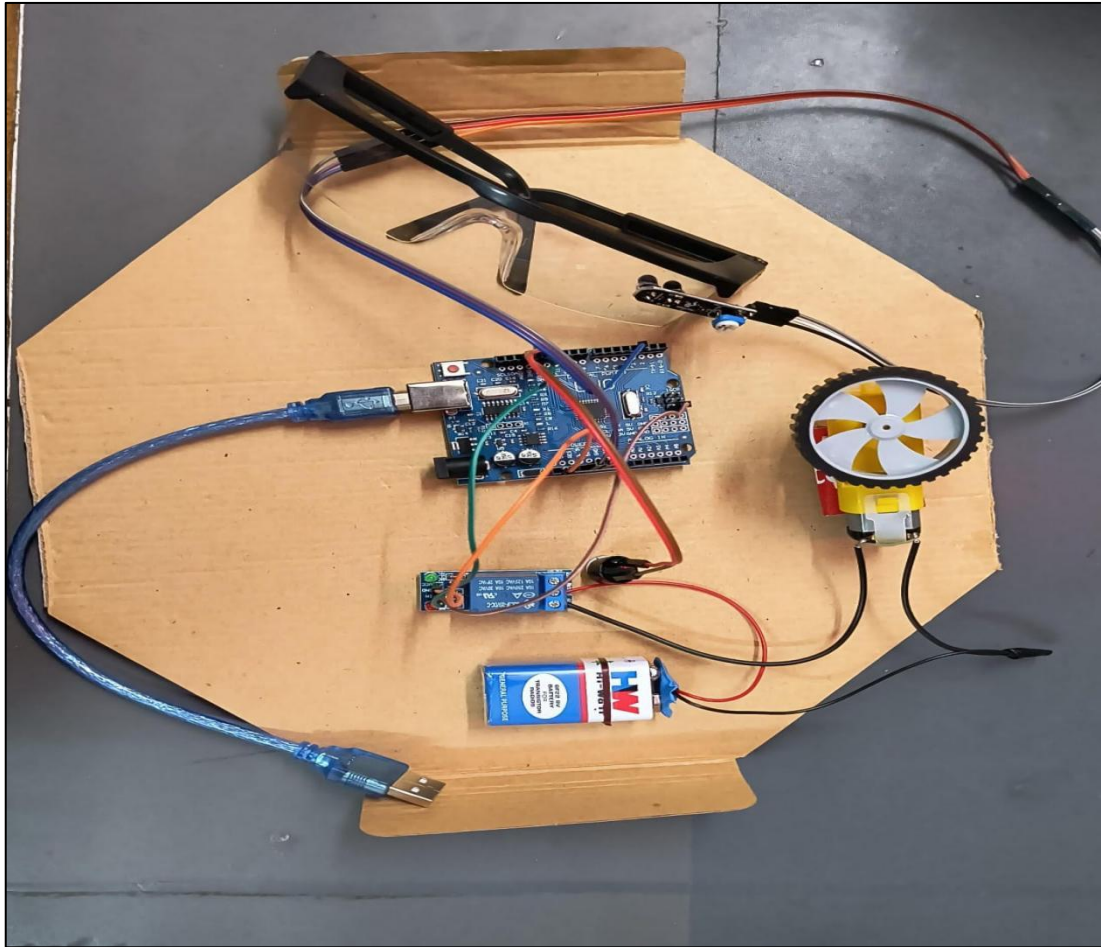
void setup() {
  pinMode(motorPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);
  pinMode(blinkPin, INPUT);
  digitalWrite(motorPin, HIGH);
}
```

```
}

void loop() {

    if(!digitalRead(blinkPin)){
        time=millis();
        while(!digitalRead(blinkPin)){
            digitalWrite(buzzerPin, LOW);
            digitalWrite(motorPin, LOW);
            delay(1000);
        }
    }
    else {
        if(TimeDelay()>=3)digitalWrite(buzzerPin, HIGH);
        if(TimeDelay()>=4)digitalWrite(motorPin, HIGH);
    }
}

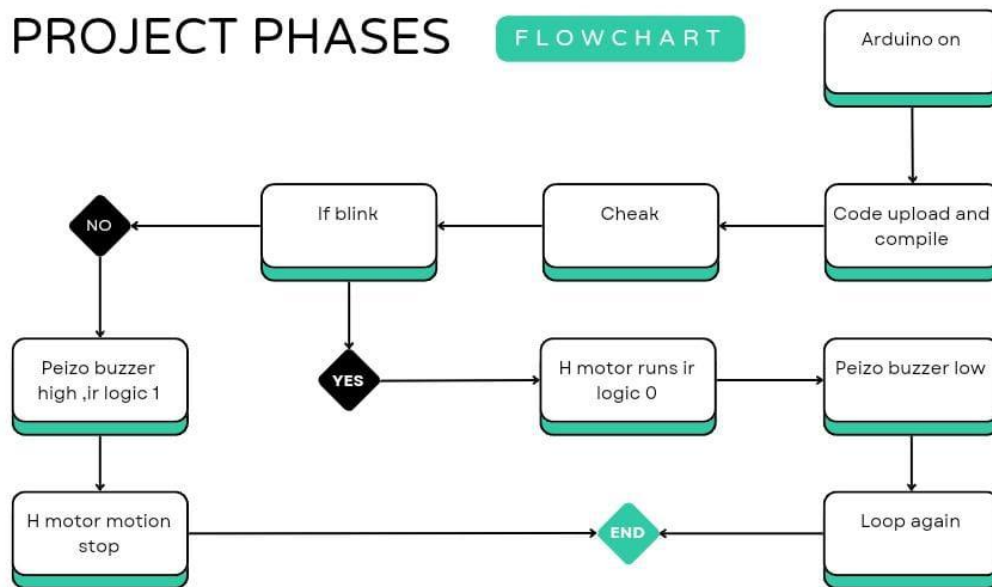
int TimeDelay(){
    long t=millis()-time;
    t=t/1000;
    return t;
}
```



:-CIRCUIT DIAGRAM:-

PROJECT PHASES

FLOWCHART



PROCEDURE

- ✧ Connect cathode of battery with hobby motor and anode with common of relay module.
- ✧ Connect another wire of hobby motor with normally closed of relay module.
- ✧ Connect 5v and ground of Arduino uno with relay module.
- ✧ Connect buzzer to pin 12 and ground.
- ✧ Connect blink sensor to 5v and ground and pin 2.
- ✧ Run and compile code.

CONCLUSION

The connection was successfully done and anti sleep alarm worked perfectly .

Working video

Press ctrl and click on link

https://drive.google.com/file/d/1kCcJM2rvE6KjCAY2x27UCMJ8hakX_5YS/view?usp=drive_link

Explanation video

https://drive.google.com/file/d/19Enz9IXtXWdgx-ZpXkAhTlCgSXLgRXcD/view?usp=drive_link

Individual contribution (self):-

In the project "Anti Sleep Alarm," my role primarily involved working on the documentation aspect. This would typically include tasks such as:

Research: Gathering information related to anti-sleep alarm systems, including their design, functionality, and potential applications.

Technical Writing: Drafting the documentation that outlines the project's objectives, methodology, system architecture, components, and operating instructions. This documentation would serve as a guide for anyone interested in understanding or replicating the project.

Editing and Formatting: Ensuring that the documentation is well-organized, clear, and concise. This may involve editing the content for grammar, spelling, and clarity, as well as formatting it to make it visually appealing and easy to navigate.

Incorporating Visuals: Adding diagrams, charts, or illustrations to enhance understanding and clarity where necessary. Visual aids can help readers grasp complex concepts more easily.

Review and Revision: Collaborating with team members to review the documentation, address feedback, and make revisions as needed to ensure accuracy and completeness.

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